

This notebook provides a hands-on introduction to Exploratory Data Analysis (EDA), demonstrating how to load data, generate descriptive statistics, and use visualization libraries like Matplotlib and Seaborn to uncover initial insights. Through various plotting techniques, such as count plots, histograms, and box plots, the lab highlights the importance of going beyond basic numerical summaries to understand data distributions and identify potential challenges like missing values or imbalanced classes.

In the visual analysis sections, the notebook explores key factors influencing passenger survival. The visualizations show that women, first-class passengers, and young children had significantly higher survival rates, supporting the historical "women and children first" narrative and indicating a socio-economic disparity in survival odds. These exercises help with the foundational skills to explore relationships between features and target variables, a critical step before proceeding to any machine learning modeling.

Initially, as most of the cells were only supposed to require running and only required coding on the exercise portion (according to notebook instructions), I used the assistance of Google Gemini to reorganize some of the code as it had all been written in a single line and was throwing errors in Google Collab. Once the code had been reorganized, I was able to continue with the module and develop the insights provided above. As I become more proficient using python, I hope to develop the skills required to make these types of adjustments without relying on Gemini as much.